



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program: Bachelor of Technology

Branch: Information Technology

Session: 2020-21

Semester : VI

Specialization: Information Technology

S.No.	Subject Code	Subject Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits
			Theory			Practical						
			End Sem.	Mid Sem. Exam.	Quiz/ Assignment	End Sem	Term work					
							Lab Work & Sessional					
L	T	P										
1.	IT-61	Machine Learning	70	20	10	30	20	150	3	1	2	5
2.	IT-62	Computer Network	70	20	10	30	20	150	3	1	2	5
3.	IT-63	Software engineering	70	20	10	-	-	100	5	-	-	5
4.	IT-64	Internet of Things	70	20	10	30	20	150	3	1	2	5
5.	IT-65	Dot Net Lab	-	-	-	30	20	50	-	-	2	1
6.	IT-66	Minor Project II	-	-	-	-	100	100	-	-	4	2
7.	IT-67	Project Seminar	-	-	-	-	50	50			2	1
		Total	280	80	40	120	230	750	14	3	14	24

*End Semester Examination will be of three hours

Practical Subject:

IT-61: Machine Learning

IT-62: Computer Network

IT-64: Internet of Things

IT-65: Dot Net Lab

IT-66: Minor Project II

IT-67: Project Seminar

1 Hr Lecture	1 Hr Tutorial	2 Hr Practical
1 Credit	1 Credit	1 Credit



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF ENGINEERING**

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

**SYLLABUS OF COURSES TO BE OFFERED
“Bachelor of Technology”**

Disclaimer: The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026

Visit us at: www.bhabhauniversity.edu.in, Mail Id info@bhabhauniversity.edu.in



NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026

Visit us at: www.bhabhauniversity.edu.in, Mail Id info@bhabhauniversity.edu.in



IT-61 Machine Learning

Unit I:- Introduction to machine learning, scope and limitations, regression, probability, statistics and linear algebra for machine learning, convex optimization, data visualization, hypothesis function and testing, data distributions, data preprocessing, data augmentation, normalizing data sets, machine learning models, supervised and unsupervised learning.

Unit II:- Linearity vs non linearity, activation functions like sigmoid, ReLU, etc., weights and bias, loss function, gradient descent, multilayer network, backpropagation, weight initialization, training, testing, unstable gradient problem, auto encoders, batch normalization, dropout, L1 and L2 regularization, momentum, tuning hyper parameters,

Unit III:- Convolutional neural network, flattening, subsampling, padding, stride, convolution layer, pooling layer, loss layer, 1x1 convolution, inception network, input channels, transfer learning, one shot learning, dimension reductions, implementation of CNN like tensor flow, keras etc.

Unit IV:- Recurrent neural network, Long short-term memory, gated recurrent unit, translation, beam search and width, Bleu score, attention model, Reinforcement Learning, RL-framework, MDP, Bellman equations, Value Iteration and Policy Iteration, Actor-critic model, Q-learning, SARSA

Unit V:- Support Vector Machines, Bayesian learning, application of machine learning in computer vision, speech processing, natural language processing etc, Case Study: Image Net Competition.

PRACTICAL:

Different problems to be framed to enable students to understand the concept learnt and get hands-on on various tools and software related to the subject. Such assignments are to be framed for ten to twelve lab sessions.

REFERENCE

1. Aurelien Geon, "Hands-On Machine Learning with Scikit-Learn and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems", Shroff/O'Reilly; First edition(2017).
2. Francois Chollet, "Deep Learning with Python", Manning Publications, 1 edition (10 January 2018).
3. Andreas Muller, "Introduction to Machine Learning with Python: A Guide for Data Scientists", Shroff/O'Reilly; First edition(2016).
4. Russell, S. and Norvig, N. "Artificial Intelligence: A Modern Approach", Prentice Hall Series in Artificial Intelligence. 2003.



IT-62 Computer Network

Unit I:- Computer Network: Definitions, goals, components, Architecture, Classifications & Types.

Layered Architecture: Protocol hierarchy, Design Issues, Interfaces and Services, Connection Oriented & Connectionless Services, Service primitives, Design issues & its functionality. ISO-

OSI Reference Model: Principle, Model, Descriptions of various layers

and its comparison with TCP/IP. Principal of physical layer: Media, Bandwidth, Data rate and Modulations

Unit-II:- Data Link Layer: Need, Services Provided, Framing, Flow Control, Error control. Data Link Layer Protocol: Elementary & Sliding Window protocol: 1-bit, Go-Back-N, Selective Repeat, Hybrid ARQ. Protocol verification: Finite State Machine Models & Petri net models. ARP/RARP/GARP

Unit III:- MAC Sub layer: MAC Addressing, Binary Exponential Back-off (BEB) Algorithm, Distributed Random Access Schemes/Contention Schemes: for Data Services (ALOHA and Slotted-ALOHA), for Local-Area Networks (CSMA, CSMA/CD, CSMA/CA), Collision Free Protocols: Basic Bit Map, BRAP, Binary Count Down, MLMA Limited Contention Protocols: Adaptive Tree Walk, Performance Measuring Metrics. IEEE Standards 802 series & their variant.

Unit IV:- Network Layer: Need, Services Provided, Design issues, Routing algorithms: Least Cost Routing algorithm, Dijkstra's algorithm, Bellman-ford algorithm, Hierarchical Routing, Broadcast Routing, Multicast Routing. IP Addresses, Header format, Packet forwarding, Fragmentation and reassembly, ICMP, Comparative study of IPv4 & IPv6

Unit V:- Transport Layer: Design Issues, UDP: Header Format, Per-Segment Checksum, Carrying Unicast/Multicast Real-Time Traffic, TCP: Connection Management, Reliability of Data Transfers, TCP Flow Control, TCP Congestion Control, TCP Header Format, TCP Timer Management. Application Layer: WWW and HTTP, FTP, SSH, Email (SMTP, MIME, IMAP), DNS, Network Management (SNMP).

LIST OF EXPERIMENTS

- 1) Study of Different Type of LAN & Network Equipment's.
- 2) Study and Verification of standard Network topologies i.e. Star, Bus, Ring etc.
- 3) LAN installations and Configurations.
- 4) Write a program to implement various types of error correcting techniques.
- 5) Write a program to Implement various types of framing methods.
- 6) Study of Tool Command Language (TCL).
- 7) Study and Installation of Standard Network Simulator: N.S-2, N.S3, OpNet, QualNet etc.
- 8) Study & Installation of ONE (Opportunistic Network Environment) Simulator for High Mobility Networks.
- 9) Configure 802.11 WLAN.
- 10) Implement & Simulate various types of routing algorithm.
- 11) Study & Simulation of MAC Protocols like Aloha, CSMA, CSMA/CD and CSMA/CA using Standard Network Simulators.
- 12) Study of Application layer protocols-DNS, HTTP, HTTPS, FTP and TelNet.



BHABHA UNIVERSITY, BHOPAL

REFERENCES

- 1) 1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks" Pearson Education.
- 2) 2 Douglas E Comer, "Internetworking With Tcp/Ip Principles, Protocols, And Architecture - Volume I" 6th Edition, Pearson Education
- 3) 3. Dimitri Bertsekas, Robert Gallager, "Data Networks", PHI Publication, Second Edition.
- 4) 4. Kaveh Pahlavan, Prashant Krishnamurthy, "Networking Fundamentals", Wiley Publication.
- 4) Uyless Black, "Computer Networks", PHI Publication, Second Edition.
- 5) Ying-Dar Lin, Ren-Hung Hwang, Fred Baker, "Computer Networks: An Open Source Approach", McGraw Hill.





IT-63 Software Engineering

Unit I:- The Software Product and Software Process

Software Product and Process Characteristics, Software Process Models: Linear Sequential Model, Prototyping Model, RAD Model, Evolutionary Process Models like Incremental Model, Spiral Model, Component Assembly Model, RUP and Agile processes. Software Process customization and improvement, CMM, Product and Process Metrics

Unit II:- Requirement Elicitation, Analysis, and Specification

Functional and Non-functional requirements, Requirement Sources and Elicitation Techniques, Analysis Modeling for Function-oriented and Object-oriented software development, Use case Modeling, System and Software Requirement Specifications, Requirement Validation, Traceability

Unit III:- Software Design

The Software Design Process, Design Concepts and Principles, Software Modeling and UML, Architectural Design, Architectural Views and Styles, User Interface Design, Function-oriented Design, SA/SD Component Based Design, Design Metrics.

Unit IV:- Software Analysis and Testing

Software Static and Dynamic analysis, Code inspections, Software Testing, Fundamentals, Software Test Process, Testing Levels, Test Criteria, Test Case Design, Test Oracles, Test Techniques, Black-Box Testing, White-Box Unit Testing and Unit Testing Frameworks, Integration Testing, System Testing and other Specialized Testing, Test Plan, Test Metrics, Testing Tools. , Introduction to Object-oriented analysis, design and comparison with structured Software Engg.

Unit V:- Software Maintenance & Software Project Measurement

Need and Types of Maintenance, Software Configuration Management (SCM), Software Change Management, Version Control, Change control and Reporting, Program Comprehension Techniques, Re-engineering, Reverse Engineering, Tool Support. Project Management Concepts, Feasibility Analysis, Project and Process Planning, Resources

Allocations, Software efforts, Schedule, and Cost estimations, Project Scheduling and Tracking, Risk Assessment and Mitigation, Software Quality Assurance (SQA). Project Plan, Project Metrics.

REFERENCES

- 1) Pankaj Jalote, "An Integrated Approach to Software Engineering", Narosa Pub, 2005
- 2) Rajib Mall, "Fundamentals of Software Engineering" Second Edition, PHI Learning
- 3) R. S. Pressman, "Software Engineering: A Practitioner's Approach", Sixth edition 2006, McGraw-Hill.
- 4) Sommerville, "Software Engineering", Pearson Education.
- 5) Richard H. Thayer, "Software Engineering & Project Management", Wiley India
- 6) Waman S. Jawadekar, "Software Engineering", TMH
- 7) Bob Hughes, M. Cotterell, Rajib Mall "Software Project Management", McGraw Hill



IT-64 Internet of Things

Unit 1:- Introduction: Definition, Characteristics of IOT, IOT Conceptual framework, IOT Architectural view, Physical design of IOT, Logical design of IOT, Application of IOT.

Unit 2:- Machine-to-machine (M2M), SDN (software defined networking) and NFV(network function virtualization) for IOT, data storage in IOT, IOT Cloud Based Services.

Unit 3:- Design Principles for Web Connectivity: Web Communication Protocols for connected devices, Message Communication Protocols for connected devices, SOAP, REST, HTTP Restful and WebSockets. Internet Connectivity Principles: Internet Connectivity, Internet based communication, IP addressing in IOT, Media Access control.

Unit 4:- Sensor Technology, Participatory Sensing, Industrial IOT and Automotive IOT, Actuator, Sensor data Communication Protocols, Radio Frequency Identification Technology, Wireless Sensor Network Technology.

Unit 5:- IOT Design methodology: Specification-Requirement, process, model, service, functional & operational view. IOT Privacy and security solutions, Raspberry Pi & Arduino devices. IOT Case studies: smart city streetlights control & monitoring.

LIST OF EXPERIMENTS

- 1) Familiarization with Arduino /Raspberry Pi and perform necessary software installation.
- 2) To interface LED/Buzzer with Arduino / Raspberry Pi and write a program to turn ON LED for 1 sec after every 2 seconds.
- 3) To interface Push button/Digital sensor (IR/LDR) with Arduino/Raspberry Pi and write a program to turn ON LED when push button is pressed or at sensor detection.
- 4) To interface DHT11 sensor with Arduino/Raspberry Pi and write a program to print temperature and humidity readings.
- 5) To interface motor using relay with Arduino/Raspberry Pi and write a program to turn ON motor when push button is pressed.
- 6) To interface OLED with Arduino/Raspberry Pi and write a program to print temperature and humidity readings on it.
- 7) To interface Bluetooth with Arduino/Raspberry Pi and write a program to send sensor data to smartphone using Bluetooth.
- 8) To interface Bluetooth with Arduino/Raspberry Pi and write a program to turn LED ON/OFF when '1'/'0' is received from smartphone using Bluetooth.
- 9) Write a program on Arduino/Raspberry Pi to upload temperature and humidity data to thing speak cloud.
- 10) Write a program on Arduino/Raspberry Pi to retrieve temperature and humidity data from thing speak cloud.
- 11) To install MySQL database on Raspberry Pi and perform basic SQL queries.
- 12) Write a program on Arduino/Raspberry Pi to publish temperature data to MQTT broker.
- 13) Write a program on Arduino/Raspberry Pi to subscribe to MQTT broker for temperature data and print it.

REFERENCES



BHABHA UNIVERSITY, BHOPAL

1. Rajkamal, "Internet of Things", Tata McGrawHill publication
2. Vijay Madiseti and Arshdeep Bahga, "Internet of things (A-Hand-on-Approach)" 1st Edition, Universal Press
3. Hakima Chaouchi "The Internet of Things: Connecting Objects", Wiley publication.
4. Charles Bell "MySQL for the Internet of things", Apress publications.
5. Francis da Costa "Rethinking the Internet of things: A scalable Approach to connecting everything", 1st edition, Apress publications 2013.
6. Donald Norris "The Internet of Things: Do-It-Yourself at Home Projects for Arduino, Raspberry Pi and BeagleBone Black", McGrawHill publication.





BHABHA UNIVERSITY, BHOPAL

IT-65 Dot Net Lab

Introduction .NET framework, features of .Net framework, architecture and component of .Net, elements of .Net.

Basic Features Of C# Fundamentals, Classes and Objects, Inheritance and Polymorphism, Operator Overloading, Structures. Advanced Features Of C# Interfaces, Arrays, Indexers and Collections; Strings and Regular Expressions, Handling Exceptions, Delegates and Events.

Installing ASP.NET framework, overview of the ASP.NET framework, overview of CLR, class library, overview of ASP.NET control, understanding HTML controls, study of standard controls, validation controls, rich controls. Windows Forms: All about windows form, MDI form, creating windows applications, adding controls to forms, handling Events, and using various Tools

Understanding and handling controls events, ADO.NET- Component object model, ODBC, OLEDB, and SQL connected mode, disconnected mode, dataset, data-reader Database controls: Overview of data access data control, using grid view controls, using details view and frame view controls, ado .net data readers, SQL data source control, object data source control, site map data source.

XML: Introducing XML, Structure, and syntax of XML, document type definition (DTD), XML Schema, Document object model, Presenting and Handling XML. xml data source, using navigation controls, introduction of web parts, using javascript, Web Services

LIST OF EXPERIMENTS

- 1) Working with call backs and delegates in C#
- 2) Code access security with C#.
- 3) Creating a COM+ component with C#.
- 4) Creating a Windows Service with C#
- 5) Interacting with a Windows Service with C#
- 6) Using Reflection in C#
- 7) Sending Mail and SMTP Mail in C#
- 8) Perform String Manipulation with the String Builder and String Classes in C#:
- 9) Using the System .Net Web Client to Retrieve or Upload Data with C#
- 10) Reading and Writing XML Documents with the XML Text-Reader/-Writer Class in C#
- 11) Working with Page using ASP.NET.
- 12) Working with Forms using ASP.NET
- 13) Data Sources access through ADO.NET,
- 14) Working with Data readers, Transactions
- 15) Creating Web Application.



BHABHA UNIVERSITY, BHOPAL

REFERENCES

1. C# for Programmers by Harvey Deitel, Paul Deitel, Pearson Education
2. Balagurusamy; Programming in C#; TMH
3. Web Commerce Technology Handbook by Daniel Minoli, Emma Minoli, TMH
4. Web Programming by Chris Bates, Wiley
5. Alex Mackey, "Introduction .NET 4.5", Wiley India
6. ASP .Net Complete Reference by McDonald, TMH.
7. ADO .Net Complete Reference by Odey, TMH





IT-66 Minor Project II

Objectives of the course Minor Project are:

- ☐ To provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses.
- ☐ To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems.
- ☐ To give students an opportunity to do something creative and to assimilate real life work situation in institution.
- ☐ To adapt students for latest development and to handle independently new situations.
- ☐ To develop good expressions power and presentation abilities in students.

The focus of the Minor Project-II is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write up i.e. detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration of the working system.

In Minor Project-II students have to make project in any of the programming language by using some kind of software and hardware which he/she has learnt in previous semesters. Students can make project individually or in a group with two-three students.

Students should make Project in the topics which will be beneficial to them in future also. Students have to Prepare file of the project which he/she will make.



BHABHA UNIVERSITY, BHOPAL

IT-67 Project Seminar

In Project Seminar students have to make a PPT of the selected topic of Project and present in front of the faculties & students. By doing this, in front of faculties & students, students communication skill will get developed & will be a good presenter. By doing this, students Project knowledge will also get enhanced.

